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Studierichting: Informatica
Oefeningenreeks 3D nr. 7.2

Bepaal de extrema van $f(x) = e^x \sin x$.

1. Eerste afgeleide

$$
\begin{aligned}
f'(x) &= \frac{d}{dx}(e^x \sin x) \\
&= \left(\frac{d}{dx}e^x\right)\sin x + e^x\left(\frac{d}{dx}\sin x\right) \\
&= e^x \sin x + e^x \cos x \\
&= e^x (\sin x + \cos x)
\end{aligned}
$$

2. Kritische punten

We zoeken $f'(x) = 0$.

$$e^x (\sin x + \cos x) = 0$$

Omdat $e^x > 0$ voor alle x geldt:

$$\sin x + \cos x = 0$$

$$\Leftrightarrow \sin x = -\cos x$$

$$\Leftrightarrow \tan x = -1$$

$$\Leftrightarrow x_k = -\frac{\pi}{4} + k\pi,\ k \in \mathbb{Z}$$

3. Tweede afgeleide

$$
\begin{aligned}
f''(x) &= \frac{d}{dx}\left(e^x(\sin x + \cos x)\right) \\
&= \left(\frac{d}{dx}e^x\right)(\sin x + \cos x) + e^x\left(\frac{d}{dx}(\sin x + \cos x)\right) \\
&= e^x(\sin x + \cos x) + e^x(\cos x - \sin x) \\
&= 2e^x \cos x
\end{aligned}
$$

Daarom:

$$f''(x_k) = 2e^{x_k}\cos x_k = 2e^{x_k}\frac{\sqrt{2}}{2}(-1)^k = \sqrt{2}e^{x_k}(-1)^k$$

4. Classificatie en specificatie

Uit $x_k = -\frac{\pi}{4} + k\pi$ onderscheiden we:

- Stel $k = 2n$ (even).

  $$x_{2n} = -\frac{\pi}{4} + 2n\pi.$$

  Dan $(-1)^{2n} = +1$, dus $f''(x_{2n}) > 0$.

  $\Rightarrow$ lokaal minimum bij

  $$x = -\frac{\pi}{4} + 2n\pi,\ n \in \mathbb{Z}$$

- Stel $k = 2n + 1$ (oneven).

  $$x_{2n+1} = -\frac{\pi}{4} + (2n+1)\pi = -\frac{\pi}{4} + 2n\pi + \pi = \frac{3\pi}{4} + 2n\pi.$$

  Dan $(-1)^{2n+1} = -1$, dus $f''(x_{2n+1}) < 0$.

  $\Rightarrow$ lokaal maximum bij

  $$x = \frac{3\pi}{4} + 2n\pi,\ n \in \mathbb{Z}$$

5. Waarden in de extrema

$$f(x_k) = e^{x_k}\sin x_k = e^{x_k}\left(-\frac{\sqrt{2}}{2}(-1)^k\right) = -\frac{\sqrt{2}}{2}(-1)^k e^{x_k}$$

- Lokaal minimum ($k = 2n$):

  $$x = -\frac{\pi}{4} + 2n\pi,\ f(x) = -\frac{\sqrt{2}}{2}\, e^{-\frac{\pi}{4}+2n\pi}.$$

- Lokaal maximum ($k = 2n + 1$):

  $$x = \frac{3\pi}{4} + 2n\pi,\ f(x) = +\frac{\sqrt{2}}{2}\, e^{\frac{3\pi}{4}+2n\pi}.$$

References